Required Capstone Checkpoint 1.1: Evaluating When Retrieval Is Required

Directions

Select one of the provided capstone scenarios and use it as the foundation for this checkpoint. You will continue developing this same scenario throughout the program. Download this worksheet and save a copy to your device before you begin working. Type your responses directly into the document, save your completed work, and upload the finished file when you are ready to submit.

Use this worksheet to:

- Analyze how an LLM performs without retrieval
- Document prompts and corresponding model outputs
- Identify limitations and patterns in model behavior
- Determine when retrieval is required
- Reflect on how retrieval quality affects reliability and user trust

1. Selected Scenario and Intended Use

Select your capstone scenario:

☐ Wikipedia Retrieval Engine

Briefly describe the kinds of questions, tasks, or user interactions your system should support.
Example considerations: ● Answering factual questions ● Comparing information across documents ● Supporting long-running conversational interactions ● Providing grounded responses with source quotations

© MIT xPRO® [current year to be added]. All rights reserved.

2. Prompt and Output Evidence

- Provide two prompts relevant to your scenario. Where possible, use prompts that test different types of information needs or tasks.
- Include the corresponding LLM outputs for each prompt.

Prompt 1 and Corresponding LLM Outputs
Which actor has won the most academy awards between from 2000 to 2020?

Response from RUN 1:

From **2000 to 2020**, the actor with the **most Academy Awards wins** is **Meryl Streep**.

She won **3 Oscars** in that period:

- **2003** — *The Hours* (Best Supporting Actress)

- **2012** — *The Iron Lady* (Best Actress)

- **2020** — *…?*

However, I should be careful here: **Meryl Streep did not win an Oscar in 2020**. So within **2000–2020 inclusive**, her Oscar wins are actually **2**:

- **2003** — *The Hours*

- **2012** — *The Iron Lady*

Among actors/actresses, **Meryl Streep** is generally the one with the most Academy Award wins in that span, with **2 wins**. If you want, I can also list the other top winners from 2000–2020.

Response from RUN 2:

If you mean **Best Actor / Best Supporting Actor acting wins at the Academy Awards** in the years **2000 through 2020**, the answer is:

Daniel Day-Lewis — he won **3** Academy Awards in that period:

- **2008** for *There Will Be Blood*

- **2013** for *Lincoln*

- (His third Oscar win overall was actually **1990s-2000s adjacent**? To be precise, his Academy Award wins are 1989, 2008, and 2013.)

© MIT xPRO® [current year to be added]. All rights reserved.

However, if you mean **all actors, including actresses**, the actors with the most wins from **2000 to 2020** are still typically **multiple people tied with 2 wins** rather than one person with more than that. For example:

- **Meryl Streep** won acting Oscars before 2000, but not during 2000–2020.
- **Cate Blanchett** won 2 total, but only 1 in that range.
- **Daniel Day-Lewis** is the standout for male actors.

So the safest answer is:

Daniel Day-Lewis has the most acting Academy Award wins between 2000 and 2020, with 2 wins in that period.

If you want, I can also give you a **full list of all actors/actresses who won multiple Academy Awards from 2000–2020**.

Prompt 2 and Corresponding LLM Outputs

Where were the 1972 Winter Olympics held?

Which country got the most medals in the 1972 Winter Olympics?

Response RUN1:

The **1972 Winter Olympics** were held in **Sapporo, Japan**.

The **Soviet Union** got the most medals at the **1972 Winter Olympics**, with **16 total medals**.

Response RUN2:

The **1972 Winter Olympics** were held in **Sapporo, Japan**.

The **Soviet Union** won the most medals at the **1972 Winter Olympics**.

© MIT xPRO® [current year to be added]. All rights reserved.

3. Analysis of Model Limitations

- Identify key limitations in the model's responses, such as hallucinations, knowledge gaps, stale information, inconsistency, vagueness, or missing information.
- Use specific prompt/output examples to support your analysis.

Key limitations and supporting prompt/output examples
The model showed several limitations across the test prompts. First, it produced factual hallucinations when a question required combining or comparing multiple pieces of information. For example, when asked which actor had won the most Academy Awards between 2000 and 2020, Run 1 answered Meryl Streep and included incorrect details about her wins. Run 2 produced a different answer and corrected itself partially, but still contained inaccurate supporting statements. The model also failed to correctly identify a person nominated across several different Oscar Awards. Second, the model had difficulty with multi-fact extraction from a larger body of text. When asked to provide three events from the year 1580, it generated events and quotations that were not supported by the source. Third, factual correctness and grounding were separate issues. In the 100-meter world record question, the underlying factual answer was correct, but the text presented as a direct Wikipedia quotation did not exactly match the article. This means a response may be factually correct while still being ungrounded at the quotation or citation level. Finally, the two runs were sometimes inconsistent with one another, especially on questions requiring aggregation or reasoning over several facts. This suggests that the model may generate a plausible answer rather than reliably determining the answer from evidence.

© MIT xPRO® [current year to be added]. All rights reserved.

4. Patterns in Model Behavior

- Describe any meaningful patterns or trends you observe across the outputs.
- Consider whether certain types of prompts produce more reliable responses than others.

Meaningful patterns or trends
A clear pattern was that straightforward, commonly stated factual questions were more reliable than questions requiring aggregation, comparison, or extraction of multiple facts. For example, both runs correctly answered where the 1972 Winter Olympics were held and which country won the most medals. These questions have a single, clearly stated answer. In contrast, performance degraded when the model had to combine information across several years, identify overlap among multiple records, or extract several facts from a larger article. The Academy Award and 1580 questions produced incorrect or hallucinated details. Another pattern was that the model often generated answers that were plausible but not sufficiently grounded. The Meryl Streep answer is a good example: she is strongly associated with Academy Award records, so the response sounded reasonable even though it did not correctly answer the question for the requested time period. Overall, the results suggest that model reliability decreases as the task moves from retrieving a single familiar fact toward retrieval plus aggregation, comparison, or reasoning over multiple facts.

© MIT xPRO® [current year to be added]. All rights reserved.

5. Determining Whether Retrieval Is Required

- Explain whether retrieval is required for your scenario and identify which observed limitations retrieval could help address.
- Justify your reasoning using evidence from the outputs.

Explanation and justification
Retrieval is required for the Wikipedia Retrieval Engine because the intended system should provide answers that are not only plausible, but grounded in the actual source material. The experiment showed that the model can answer some simple factual questions correctly from its internal knowledge, such as the location of the 1972 Winter Olympics. However, it became unreliable when questions required exact quotations, comparison across multiple years, or aggregation of several facts. Retrieval could address these limitations by supplying the model with the relevant Wikipedia passages before generation. For questions involving comparisons or aggregation, the retrieval system would need to retrieve all of the relevant records rather than only one highly similar passage. Retrieval is especially important when the user asks for an exact quotation or source-supported answer. Even when the model knew the correct 100-meter world-record fact, it did not reproduce the source quotation faithfully. Therefore, retrieval is necessary not only to improve factual accuracy but also to improve grounding and citation fidelity.

© MIT xPRO® [current year to be added]. All rights reserved.

6. Retrieval Requirements

- Describe the types of information that would need to be retrieved to improve reliability, grounding, and response quality. Consider whether the information is proprietary, frequently changing, highly specific, or otherwise unavailable to the model without retrieval.
- Explain when retrieval would or would not be necessary.

Information that would need to be retrieved
The system would need to retrieve the specific Wikipedia content required to answer the question, including relevant article passages, dates, names, tables, lists, and source text used for quotations. For simple factual questions, one relevant passage may be sufficient. For questions requiring comparison or aggregation, the system may need to retrieve multiple relevant passages or records. For example, determining whether one person was nominated at the 68th, 70th, and 74th Academy Awards requires information from all three ceremonies so that the results can be compared. When the user requests an exact quotation, the system should retrieve the original text rather than rely on the language model to reconstruct it from memory. Because Wikipedia content can also change over time, retrieval provides access to the version of the information available to the system at query time rather than relying only on information learned during model training.

When retrieval would or would not be necessary
Retrieval may be less necessary for casual questions involving stable and widely known facts when the user does not require source verification. For example, a general conversational answer about where the 1972 Winter Olympics were held may be answered correctly without retrieval.

© MIT xPRO® [current year to be added]. All rights reserved.

7. Reflection and Next Steps

Explain how the precision, relevance, and accuracy of retrieved information affect output quality, system reliability, and user trust. Consider how retrieval quality might improve or degrade system performance. What could happen if the retrieved information is incomplete, loosely related, outdated, or inaccurate?

Explanation

Retrieval quality directly affects the quality and reliability of the generated answer. Relevant and accurate retrieved information gives the model evidence it can use to produce a grounded response. However, retrieval by itself does not guarantee correctness.

If the retrieved information is incomplete, the model may answer from only part of the available evidence and reach the wrong conclusion. This is particularly important for comparison or aggregation questions. For example, a question about Academy Award nominations across three different years cannot be answered reliably if the system retrieves information for only one or two of those years.

If retrieved passages are only loosely related to the question, the model may combine relevant and irrelevant information and produce a plausible but unsupported answer. Outdated or inaccurate retrieved content can also cause the model to confidently generate an incorrect response.

The experiment also showed that factual correctness and grounding are different dimensions of quality. A response may contain the correct fact but still cite or quote the source inaccurately. For a system that promises Wikipedia-grounded answers, inaccurate quotations or citations could reduce user trust even when the underlying fact is correct.

Therefore, retrieval should be evaluated not only on whether it finds something relevant, but on its precision, completeness, freshness, and ability to provide sufficient evidence for the requested task.

© MIT xPRO® [current year to be added]. All rights reserved.

8. Retrieval Requirements

- Reflect on what you learned about LLM limitations, common failure modes, and retrieval needs from this activity.
- Describe what you would investigate or improve next as you continue developing your capstone system.

Reflection
This activity showed me that LLMs can perform well on simple factual questions while becoming less reliable when a task requires comparison, aggregation, exact source wording, or reasoning across multiple pieces of evidence. It also showed that asking the same question multiple times can help identify potentially unreliable behavior. If the model produces different answers across runs, that inconsistency can be an early signal that retrieval or additional grounding may be needed, especially when ground truth is not readily available. However, consistency alone does not guarantee correctness, since a model may repeatedly produce the same incorrect answer. The next step is to implement RAG calls to the LLM and compare different retrieval approaches, such as BM25 lexical search and vector search. One of the key challenges will be retrieving the right and sufficiently complete information to support the answer. I would compare how different retrieval methods perform on simple factual questions, exact quotation requests, and questions that require combining information from multiple passages or documents. Originally, I was considering using research papers as the primary source for my capstone project rather than Wikipedia. However, working through this exercise made me realize that building an effective RAG system is not only a retrieval problem; it also requires being able to evaluate whether the retrieved information is complete, relevant, and appropriate for the question. For research papers, that evaluation often requires subject-matter expertise to curate the corpus, interpret the content correctly, and recognize when important evidence is missing or misleading. I do not necessarily have enough domain expertise across a broad set of research topics to make those judgments reliably. Wikipedia therefore seems like a better fit for this capstone because the content is more general, easier for me to validate independently, and still provides enough complexity to test retrieval, grounding, aggregation, and citation quality.

© MIT xPRO® [current year to be added]. All rights reserved.

Supporting Materials

Use this section to list, describe, or link to any supporting materials included with your submission.

Supporting material	Location, link, or note
checkpoint_1_1_responses.run1.log	Raw LLM responses from Run 1 used in the analysis https://github.com/evbaron/rag_and_context_engineering/blob/main/wikipedia_retrieval_engine/Capstone%20Checkpoint%201.1/checkpoint_1_1_responses.run1.log
checkpoint_1_1_responses.run2.log	Raw LLM responses from Run 2 used in the analysis https://github.com/evbaron/rag_and_context_engineering/blob/main/wikipedia_retrieval_engine/Capstone%20Checkpoint%201.1/checkpoint_1_1_responses.run2.log
AI Assistance Disclosure: I used ChatGPT to help review the model outputs, organize my observations, and refine the wording of my written analysis. The prompts, experimental runs, interpretations, and final conclusions were reviewed and selected by me.	

End of Required Capstone Checkpoint 1.1

Return to the assignment page in the learning platform to upload and submit your completed capstone checkpoint and complete any remaining items in this module.

© MIT xPRO® [current year to be added]. All rights reserved.